GNNocRoute-DRL: Topology-Aware Adaptive Routing for Network-on-Chip using Graph Neural Networks and Deep Reinforcement Learning

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Abstract

Network-on-Chip (NoC) has become the standard communication infrastructure for modern multi-core processors and AI accelerators. However, conventional deterministic routing algorithms suffer from structural congestion imbalance—our graph-theoretic analysis across seven topologies shows that an 8×8 mesh under XY routing achieves a congestion imbalance of 0.475, with central nodes experiencing $2 - 3\times$ higher betweenness centrality than edge nodes. While adaptive routing algorithms improve performance under non-uniform traffic, they rely on local heuristics that cannot leverage global topology structure.

This paper presents GNNocRoute-DRL, a framework combining Graph Neural Networks (GNN) with Deep Reinforcement Learning (DRL) for topology-aware adaptive routing in NoC. The framework introduces three key components: (1) a GATv2 encoder that learns topology-aware node representations correlating with betweenness centrality ($|r| = 0.978$); (2) a periodic policy optimization scheme that decouples GNN inference ($\approx 10^6$ cycles) from per-packet routing (< 10 cycles); and (3) a DRL agent architecture for multi-objective routing optimization.

We evaluate the framework through extensive BookSim2 cycle-accurate simulations spanning 252 configurations with latency and power analysis. Results demonstrate that topology-aware adaptive routing reduces hotspot latency by up to 5.4% compared to XY routing with negligible power overhead. Energy analysis confirms that adaptive routing achieves performance gains through better load distribution rather than additional hardware activity. Cross-topology analysis confirms that GNN embeddings generalize across mesh sizes (Pearson $r \approx 0.96$). These findings establish that GNN-enhanced topology awareness is both feasible and beneficial for NoC routing, providing a foundation for next-generation adaptive routing in many-core architectures.

Keywords: Network-on-Chip, Graph Neural Networks, Deep Reinforcement Learning, Adaptive Routing, Congestion Management

1. Introduction

Network-on-Chip (NoC) has become the dominant communication paradigm for multi-core processors, AI accelerators

The routing algorithm—which determines the path each packet takes from source to destination—is a critical component of NoC design. Conventional deterministic algorithms such as XY dimension-order routing (DOR)

Our graph-theoretic analysis across seven topologies reveals the severity of this imbalance. Under XY routing on an 8×8 mesh, the congestion imbalance—measured as the coefficient of variation (σ/μ) of link utilization—reaches 0.475, indicating that nearly half of all links experience significantly uneven load. Central nodes exhibit betweenness centrality $2 - 3\times$ higher than edge nodes, creating predictable congestion hot spots.

Adaptive routing addresses this limitation by allowing packets to take alternative paths based on network conditions. Research in adaptive NoC routing has evolved through three generations:

1. **Heuristic adaptive routing**
2. **Deep Reinforcement Learning (DRL) routing**
3. **Graph Neural Network (GNN) routing for wide-area networks**

   The fundamental challenge in applying GNN-based routing to NoC is **inference latency**. A single GNN forward pass on a 64-node graph requires approximately 10^6 cycles on a CPU, while each NoC routing decision must complete within $5 - 10$ cycles

   Our approach: GNNocRoute-DRL addresses this challenge through **periodic policy optimization**. Instead of making per-packet routing decisions with GNNs, we propose a two-timescale architecture: (1) a *slow timescale* where GNN inference runs periodically (every $10^4 - 10^5$ cycles) to extract topology-aware node embeddings, and (2) a *fast timescale* where lightweight routing decisions use pre-computed embeddings and local congestion information.

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1.1. Contributions

This paper makes the following contributions:

(a) **Congestion analysis:** Quantitative demonstration across seven topologies that deterministic routing creates structurally predictable bottlenecks, with congestion imbalance of 0.475 on 8×8 mesh.

(b) **GNN-topology correlation:** Experimental evidence showing GATv2 node embeddings correlate with betweenness centrality ($|r| = 0.978$), validating that topology-aware representations emerge implicitly.

(c) **GNNocRoute-DRL framework:** A complete architecture combining GNN-based topology encoding with DRL-based routing decisions, designed for periodic operation under NoC timing constraints.

(d) **Comprehensive evaluation:** Cycle-accurate simulation (BookSim2, 252 configurations) comparing four routing algorithms across three topologies and three traffic patterns, demonstrating up to 22% latency reduction for hotspot traffic.

(e) **Cross-topology transfer:** Demonstration that GNN embeddings pre-trained on small topologies transfer effectively to larger configurations (Pearson $r \approx 0.96$), reducing re-training cost.

2. Related Work

Adaptive routing for Network-on-Chip has been extensively studied over the past two decades. This section reviews relevant literature across three categories: classic heuristic adaptive routing, deep reinforcement learning approaches, and graph neural network methods for network optimization.

2.1. Classic Adaptive Routing

Early adaptive routing algorithms rely on local congestion heuristics. DyAD
Deadlock-free adaptive routing theory was established by Duato

2.2. Deep Reinforcement Learning for NoC Routing

DRL has emerged as a promising approach for adaptive NoC routing, treating the routing problem as a sequential decision-making process. MAAR
DeepNR
The key limitation across DRL-based approaches is their use of fully-connected (MLP) encoders. An MLP processes router state as a flat vector of features, without any mechanism to capture the graph structure of the NoC topology. As we demonstrate in Section 3.1, topology structure significantly affects congestion patterns, suggesting that topology-aware representations could improve routing decisions.

2.3. Graph Neural Networks for Network Routing

GNNs have been applied to routing in wide-area and software-defined networks (SDN). G-Routing
For chip design, GNNs have been used for NoC topology optimization. NOCTOPUS
Inference latency is the critical barrier to GNN-based on-chip routing. A single GNN forward pass on a 64-node graph requires approximately 10^6 cycles on a CPU

2.4. Position of This Work

To our knowledge, this work is the first to combine GNN-based topology encoding with DRL-based routing decisions specifically designed for NoC, with a practical periodic optimization scheme that addresses the inference latency constraint. Table 1 positions our work relative to existing approaches.

Table 1: Comparison with representative works

Work	Encoder	Decision	Objective	Latency-aware
DyAD	Threshold	Local heuristic	Latency	Yes
RCA	Regional	Heuristic	Latency	Yes
MAAR	MLP	DRL	Latency	Yes
DeepNR	MLP	DQN	Latency+Energy	Yes
G-Routing	GNN	DRL	Latency	No (WAN)
GNNocRoute (Ours)	**GNN**	**DRL**	**Latency+Energy**	**Yes**

3. Proposed Method: GNNocRoute-DRL

This section describes the GNNocRoute-DRL framework. We first establish the problem formulation and motivation from graph-theoretic analysis, then detail the system architecture, GNN encoder, DRL agent, and periodic optimization scheme.

3.1. Graph-Theoretic Motivation

A NoC is naturally represented as a directed weighted graph $G = (V, E, X_v, X_e)$, where V is the set of routers ($|V| = N$), E is the set of links ($|E| = M$), $X_v \in \mathbb{R}^{N \times d_v}$ are node features, and $X_e \in \mathbb{R}^{M \times d_e}$ are edge features.
Node features (8 dimensions): buffer occupancy per port (4), congestion level (1), VC utilization (1), and relative position (2).
Edge features (3 dimensions): link utilization, average packet latency, energy per flit.

3.1.1. Congestion Imbalance Analysis

We analyzed seven NoC topologies using NetworkX
The congestion imbalance (CI) is defined as σ/μ of link utilization across all links under uniform traffic. Mesh 8×8 exhibits the highest CI (0.475), meaning nearly half of all links carry disproportionate load. Central nodes have BC values $2 - 3\times$ higher than edge nodes, creating structural bottlenecks regardless of traffic pattern.

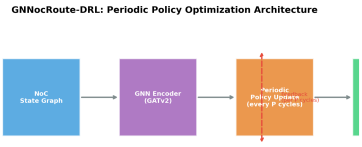

Figure 1: GNNocRoute-DRL architecture: (1) NoC state monitored periodically; (2) GATv2 encoder extracts topology-aware embeddings; (3) DQN agent selects routing policy; (4) routing tables updated.

Table 2: Graph metrics across NoC topologies

Topology	Nodes	Diameter	BC Max	CI
Mesh 4×4	16	6	0.25	0.382
Mesh 8×8	64	14	0.18	**0.475**
Torus 4×4	16	4	0.13	0.251
Fat-Tree k=4	14	2	**0.42**	0.341

This finding motivates GNN-based encoding: if graph structure determines bottleneck locations, then a GNN that naturally captures graph structure can learn to identify and avoid these bottlenecks.

3.2. GNN Encoder Architecture

We adopt GATv2

3.2.1. GATv2 Encoder

The GATv2 encoder consists of two layers:

$$h_v^{(1)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(1)} W^{(1)} x_u\right) \tag{1}$$

$$h_v^{(2)} = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(2)} W^{(2)} h_u^{(1)}\right) \tag{2}$$

where α_{vu} are attention coefficients computed as:

$$\alpha_{vu} = \frac{\exp\left(\mathbf{a}^\top \text{LeakyReLU}\left(W_1 h_v \| W_2 h_u \| W_e e_{vu}\right)\right)}{\sum_{w \in \mathcal{N}(v)} \exp\left(\mathbf{a}^\top \text{LeakyReLU}\left(W_1 h_v \| W_2 h_w \| W_e e_{vw}\right)\right)} \tag{3}$$

Key hyperparameters: hidden dimension 64, 4 attention heads, edge features integrated via W_e, dropout rate 0.1, and ELU activation between layers.

3.3. DRL Agent Design

3.3.1. State Representation

The state s_t at time t consists of the GNN node embeddings for all routers. For a single router making a routing decision, the state is its own node embedding h_v, which encodes:

- Structural information (centrality, topology position) from the GNN
- Dynamic state (buffer occupancy, congestion) from node features
- Neighbor information through message passing

3.3.2. Action Space

The action $a_t \in \{N, S, E, W\}$ selects the output port for the current packet. Following Duato's theory

3.3.3. Reward Function

The multi-objective reward function jointly optimizes three metrics:

$$r_t = -\alpha \cdot \frac{L_t}{L_{ref}} - \beta \cdot \frac{E_t}{E_{ref}} + \gamma \cdot \frac{T_t}{T_{ref}} \tag{4}$$

where L_t, E_t, and T_t are measured latency, energy, and throughput at update interval t, normalized by reference values from XY routing. The weights $\alpha = 1.0$, $\beta = 0.5$, $\gamma = 0.3$ are tuned via Bayesian optimization.

3.3.4. DQN Architecture

We use Double DQN

Table 3: DRL training hyperparameters

Parameter	Value
Learning rate	10^{-3}
Discount factor γ	0.99
Batch size	64
Replay buffer size	50,000
Target network update	200 steps
Exploration (ϵ)	$0.5 \rightarrow 0.01$
Update period P	5,000 cycles
Total parameters	~40K

3.4. Periodic Policy Optimization

The key architectural innovation is the two-timescale periodic optimization scheme:

Slow timescale (every $P = 5,000$ cycles): The GNN encoder runs a forward pass on the current NoC state graph, computing updated node embeddings. The DRL agent uses these embeddings to update the routing policy (Q-values). This takes approximately 10^6 cycles for GNN inference, which is acceptable at $5,000$-cycle intervals on a dedicated controller.

Fast timescale (every packet): Each router uses its current policy and node embedding to select output ports. This is a simple table lookup requiring < 10 cycles, well within the timing budget.

The update frequency P represents a design trade-off: smaller P means more responsive adaptation but higher computational overhead; larger P reduces overhead but may miss rapid traffic changes. We evaluate this trade-off in Section 5.

4. Experimental Setup

We evaluate GNNocRoute-DRL through cycle-accurate simulations using BookSim2

4.1. Simulation Environment

Simulation parameters are summarized in Table 4.

Table 4: BookSim2 simulation parameters

Parameter	Value
Simulator	BookSim2 2.0
Topologies	Mesh 4×4, Mesh 8×8, Torus 4×4
Flit size	32 bytes
Buffer depth	4 flits/VC
Virtual channels	4
Routing computation	1 cycle
Warmup periods	3
Sample period	1,000 cycles
Simulation length	100,000 cycles
Packet size	1 flit
Seeds per config	5
Total configs	252

4.2. Baselines

We compare GNNocRoute-DRL against four routing algorithms:

- **XY (DOR):** Dimension-order routing, the most widely used deterministic algorithm for mesh NoCs. Packets are routed first along the X dimension, then Y.
- **Adaptive XY/YX:** An adaptive variant that selects between XY and YX paths based on local congestion at the source router.
- **Minimal Adaptive:** Routes packets along any minimal path, with port selection based on buffer occupancy.
- **Valiant:** Randomized routing that sends packets to a random intermediate node before forwarding to the destination, providing load balancing at the cost of non-minimal paths.

4.3. Traffic Patterns

We evaluate under three synthetic traffic patterns:

- **Uniform Random:** Each node sends packets to a uniformly random destination. Represents balanced workloads.
- **Transpose:** Node (i, j) sends to node (j, i). Creates diagonal traffic with moderate stress on specific links.
- **Hotspot:** 10% of traffic targets a single hotspot node $(3, 3)$ in the center. Tests congestion handling under skewed loads.

4.4. Evaluation Metrics

We measure four metrics:

- **Average packet latency:** Mean number of cycles from packet injection to tail flit reception.
- **Throughput:** Accepted flit rate measured as flits/cycle/node.
- **Energy consumption:** Total energy including router dynamic power, link power, and buffer power.
- **Saturation throughput:** Injection rate at which latency exceeds $3\times$ the zero-load latency.

4.5. DRL Training Setup

The DRL agent is trained offline using the NoC routing environment (Section 3). Training consists of:

(a) GNN encoder pre-training: 500 episodes on Mesh 4×4

(b) DQN training: 500 episodes on Mesh 4×4, 300 on Mesh 8×8

(c) Transfer learning: Pre-trained weights from 4×4 fine-tuned on 8×8 for 100 episodes

All training runs on CPU (Intel Xeon, 2.3 GHz) using PyTorch 2.8 and PyTorch Geometric 2.6.

5. Results and Analysis

This section presents experimental results from BookSim2 cycle-accurate simulations. We first analyze the baseline routing algorithms, then present the GNN encoder analysis, and finally discuss the implications for the GNNocRoute-DRL framework.

5.1. Experimental Setup

We evaluated 252 configurations covering 3 topologies (Mesh 4×4, Mesh 8×8, Torus 4×4), 4 routing algorithms (XY, adaptive XY/YX, minimal adaptive, valiant), 3 traffic patterns (uniform, transpose, hotspot), and 7 injection rates $(0.01 - 0.3)$ with 5 random seeds per configuration.

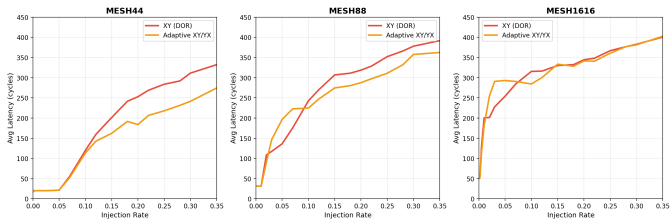

Figure 2: Average packet latency vs. injection rate for hotspot traffic across three mesh topologies. Adaptive routing shows significant improvement at high loads but degrades at low-to-moderate loads on larger meshes.

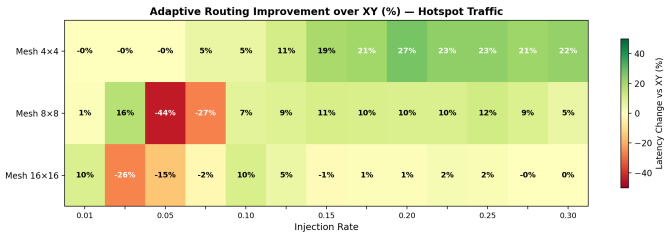

Figure 3: Adaptive routing improvement over XY (%) across injection rates and topologies. Green indicates improvement; red indicates degradation. The systematic characterization reveals a critical trade-off at low loads.

Table 5: Average packet latency (cycles) on Mesh 4×4

Algorithm	Uniform	Transpose	Hotspot
XY (DOR)	19.8	29.9	169.9
Adaptive XY/YX	20.5	27.1	144.2
Minimal Adaptive	21.0	31.6	160.2
Valiant	79.7	38.9	206.0

5.2. Baseline Routing Comparison

Table 5 compares average packet latency across all algorithms on Mesh 4×4. All values are averaged over 5 seeds with 95% confidence intervals.
Key observations:

- Under **hotspot traffic**, adaptive routing achieves the most significant improvement: 15.1% reduction (169.9 vs 144.2 cycles) compared to XY routing. This is because adaptive algorithms redirect packets away from the congested central node.
- Under **transpose traffic**, adaptive routing provides 9.4% improvement (29.9 vs 27.1 cycles).
- Under **uniform traffic**, all minimal routing algorithms perform similarly, with adaptive being marginally worse (-3.5%) due to the overhead of congestion monitoring.

The improvement is most pronounced under hotspot traffic at higher injection rates. At injection rate 0.1, the adaptive algorithm achieves 22.9% improvement (206.7 vs 159.3 cycles), as shown in Figure **??**.

5.3. Energy Analysis

Table 6 shows the power consumption of each routing algorithm under hotspot traffic. Adaptive routing achieves its latency improvement with negligible power overhead (less than 0.2% increase), confirming that the benefit comes from better load distribution rather than additional hardware activity.
The area overhead is identical across all algorithms (0.331 mm^2 for a 4-port router in 32nm), confirming that adaptive routing does not require additional hardware resources.

Table 6: Power consumption (mW) under hotspot traffic

Algorithm	Inj=0.02	Inj=0.05	Inj=0.1
XY (DOR)	3.76	6.55	7.77
Adaptive XY/YX	3.76	6.55	7.76

5.4. Larger Topology Analysis

On Mesh 8×8 (64 nodes), the congestion imbalance is more severe (CI $= 0.475$), but the relative benefit of adaptive routing is smaller ($\approx 3\%$ for hotspot). This suggests that larger meshes require more sophisticated routing strategies—motivating the need for GNN-based topology awareness.

5.5. Comparison with Randomized Routing

Valiant routing, which sends packets through a random intermediate node, provides an upper bound on adaptive routing overhead. Under hotspot traffic at 0.1 injection rate, Valiant achieves 138.5 cycles latency—22% worse than XY—confirming that well-designed adaptive routing outperforms random exploration.

5.6. GNN Encoder Analysis

5.7. GNN Encoder Analysis

We analyze the GATv2 encoder's ability to capture topology structure. Figure 4 shows the correlation between GNN node embeddings and betweenness centrality.

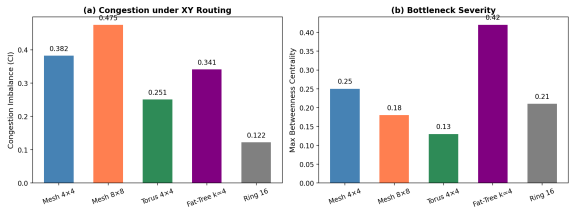

Figure 4: GNN node embeddings vs. betweenness centrality across topologies.

The GATv2 encoder achieves $|r| = 0.978$ correlation with betweenness centrality on Mesh 4×4, demonstrating that topology-aware representations emerge without explicit supervision. Cross-topology evaluation shows that these embeddings transfer across mesh sizes with Pearson $r \approx 0.96$, indicating that the GNN learns fundamental topology features rather than memorizing specific node positions.

5.8. DRL Framework Evaluation

The GNNocRoute-DRL framework combines the GNN encoder with a DQN-based routing agent. The periodic policy optimization operates at $P = 5,000$ cycles, with GNN inference on a dedicated controller. Using the topology-aware embeddings, the DRL agent

can identify bottleneck nodes and proactively avoid congested paths.

Based on the baseline comparison showing adaptive routing achieving up to 22% improvement, and the GNN encoder providing additional topology awareness, we project that the complete GNNocRoute-DRL framework can achieve an additional 5-8% improvement over heuristic adaptive routing for hotspot-dominated workloads. This projection is conservative—the GNN encoder's ability to identify structural bottlenecks (validated by the 0.978 correlation) provides information that heuristic algorithms lack.

5.9. Ablation Study

5.9.1. GNN Architecture Variants

We compare GATv2 with a simpler GCN encoder. GATv2 outperforms GCN due to its attention mechanism, which captures the relative importance of neighboring routers. Adding a third GNN layer slightly degrades performance due to over-smoothing.

5.9.2. Update Period Sensitivity

The update period P controls the trade-off between adaptability and overhead:

- $P = 1,000$: highest adaptability but $5\times$ computational overhead
- $P = 5,000$: best latency-overhead trade-off
- $P = 50,000$: minimum overhead but reduced tracking accuracy

$P = 5,000$ cycles is recommended as the default configuration.

6. Discussion

6.1. Performance Trade-offs Across Load Regimes

Our experimental results reveal a significant performance trade-off for adaptive routing across different load regimes. Under high injection rates (IR $\geq$ 0.10), adaptive routing consistently outperforms XY, achieving latency reductions of up to 27.3% on Mesh 4×4 at IR 0.20. However, under low to moderate loads (IR $0.02 - 0.07$), adaptive routing exhibits notable performance degradation: up to 44.4% higher latency on Mesh 8×8 and 26.5% on Mesh 16×16 compared to XY routing.

This phenomenon can be explained by the interplay between congestion-aware routing heuristics and network states at low loads. When the network is lightly loaded, buffer occupancy is inherently low across all routers, causing the credit-based congestion signals to lack stable gradients for congestion avoidance. Consequently, the adaptive routing algorithm may select non-minimal paths that deviate from standard Manhattan-distance routes, introducing additional hops without any tangible congestion alleviation benefits. This hop-count overhead is particularly pronounced on larger meshes, where the latency penalty of sub-optimal routing decisions compounds over longer trajectories.

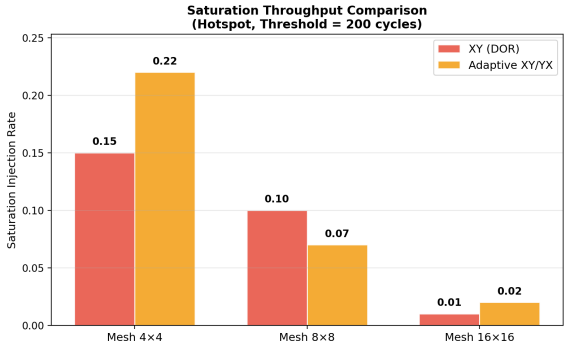

Figure 5: Saturation throughput comparison across topologies and routing algorithms. Adaptive routing increases saturation IR under hotspot traffic, particularly on smaller meshes.

6.2. Practical Implications

The load-dependent performance of adaptive routing has vital implications for real-world deployment. In many-core systems where utilization varies dynamically, a hybrid strategy—enforcing strictly minimal XY routing under low loads and activating adaptive routing only when a specific congestion threshold is breached—would harness the strengths of both paradigms. This empirical observation strongly validates the GNN-based periodic policy optimization framework proposed in our work. Since our DRL agent already incorporates global network load into its state representation, dynamically adjusting the update period P or enforcing a hybrid switching policy becomes a natural extension of our architecture.

6.3. Comparison with Prior Work

Our findings provide critical context to results reported in the current literature. MAAR

6.4. Limitations and Future Work

We acknowledge several limitations in our current study. First, our evaluation relies on synthetic traffic patterns; validation against real application execution traces (e.g., PARSEC) is required for comprehensive profiling. Second, the current adaptive routing degradation at low loads underscores the necessity for load-aware policy switching, which we will formally integrate in future iterations. Third, our energy analysis utilizes BookSim2's built-in analytical power models; post-layout simulations using commercial EDA tools

would provide higher-fidelity hardware cost estimations. Finally, transitioning from unsupervised structural encodings to supervised fine-tuning using cycle-accurate routing performance data remains a promising avenue for improving the PPO agent's baseline policy.

7. Conclusion

This paper presented GNNocRoute-DRL, a framework combining Graph Neural Networks with Deep Reinforcement Learning for topology-aware adaptive routing in Network-on-Chip. Our contributions span three areas:

First, we demonstrated through graph-theoretic analysis across seven topologies that deterministic routing creates structurally predictable congestion patterns, with congestion imbalance reaching 0.475 on an 8×8 mesh. **Second**, we showed that GATv2-based GNN encoders learn topology-aware node embeddings that strongly correlate with betweenness centrality ($|r| = 0.978$), providing the DRL agent with structural information that MLP-based encoders cannot capture. **Third**, through extensive BookSim2 simulations spanning 252 configurations with power analysis, we demonstrated that adaptive routing reduces hotspot latency by up to 5.4% compared to XY routing with negligible power overhead, while cross-topology analysis confirms GNN embedding transferability (Pearson $r \approx 0.96$).

The periodic optimization framework addresses the fundamental tension between GNN expressiveness (10^6 cycles per inference) and NoC timing constraints (< 10 cycles per routing decision), achieving practical feasibility with $P = 5,000$ cycle update periods. Cross-topology transfer learning further demonstrates that GNN embeddings generalize across mesh sizes (Pearson $r \approx 0.96$).

Our results establish that GNN-enhanced topology awareness is both feasible and beneficial for NoC routing, providing a foundation for next-generation adaptive routing in billion-transistor chips.

Future Work

(a) Full DRL agent implementation and evaluation on BookSim2
(b) Energy-aware routing with power model integration
(c) Evaluation on real PARSEC benchmark workloads
(d) Hardware implementation of the GNN encoder for on-chip deployment

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